

H M A Mohit Chowdhury

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I do research in the intersection of Deep Learning and Genomics, with a focus on large-scale time-series DNA-seq data. My research bridges the gap between complex biological sequences and state-of-the-art AI, utilizing Self-Supervised Learning, Computer Vision, Transformers, and Diffusion models for high-fidelity forecasting and representation learning. Beyond the bench, I am a builder—designing scalable, cloud-native AI systems in Python to translate high-impact research into robust healthcare applications.

Education

Ph.D. in Computer Science and Engineering

University of North Texas

2025 - 2027 (Expected)

University of Colorado Colorado Springs (Transferred to UNT)

2022 - 2025

M.Sc. in Computer Science

GPA: 3.867

University of Colorado Colorado Springs

2024

B.Sc. in Computer Science and Engineering

GPA: 3.71

Ahsanullah University of Science and Technology

2017

Scientific Research (Google Scholar)

- **Chowdhury, H. M. A. M.**, Oluwadare, O. *HiCInterpolate: 4D Spatiotemporal Interpolation of Hi-C Data for Genome Architecture Analysis*. bioRxiv (2026), <https://doi.org/10.64898/2026.02.06.704438>
- **Chowdhury, H. M. A. M.**, Oluwadare, O. *EmbedTAD Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data*. Communications Biology 9, 7 (2026), <https://doi.org/10.1038/s42003-025-09224-z>
- **Chowdhury, H. M. A. M.**, Fuller, M., & Oluwadare, O. *Robin: an advanced tool for comparative loop caller result analysis leveraging large language models*. NAR Genomics and Bioinformatics 8(1), 2026, <https://doi.org/10.1093/nargab/lqag009>
- Pinchuk, D., **Chowdhury, H. M. A. M.**, Pandeya, A., & Oluwadare, O. *HiCForecast: dynamic network optical flow estimation algorithm for spatiotemporal Hi-C data forecasting*. Bioinformatics 41(2), 2025, <https://doi.org/10.1093/bioinformatics/btaf030>
- Rohit, M., **Chowdhury, H. M. A. M.**, & Oluwadare, O. *ScHiCAtt: Hi-C Super Resolution Using Attention Methods for Single Cell*. Computational and Structural Biotechnology Journal 27, 2024, <https://doi.org/10.1016/j.csbj.2025.02.031>
- **Chowdhury, H. M. A. M.**, Boulton, T., & Oluwadare, O. *Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness*. BMC Bioinformatics 25(1), 2024, <https://doi.org/10.1186/s12859-024-05713-w>
- Houchens, D., **Chowdhury, H. M. A. M.**, & Oluwadare, O. *coiTAD: Detection of Topologically Associating Domains Based on Clustering of Circular Influence Features from Hi-C Data*. Genes 15(10), 2024, <https://doi.org/10.3390/genes15101293>
- Akpoki, V., **Chowdhury, H. M. A. M.**, Olowofila, S., Nusrat, R., & Oluwadare, O. *CNNsplice: Robust models for splice site prediction using convolutional neural networks*. Computational and Structural Biotechnology Journal 21, 2023, <https://doi.org/10.1016/j.csbj.2023.05.031>

Professional Service

- Reviewer: BMC Bioinformatics, Scientific Reports, Computational and Structural Biotechnology Journal (CSBJ), Genome Biology, BioData Mining, IEEE International Conference on Bioinformatics and Biomedicine (BIBM), and Analytical Letters
- Mentor: UR2PHD Program, University of North Texas (Fall 2025 – Spring 2026)

Skills

- **AI / ML:** Deep Learning, Time-Series, Spatiotemporal Modeling, Transformer Architecture, Diffusion Architecture, CNNs, Graph Representation, Clustering.

- **Programming Languages:** Python, Java, C#, C, JavaScript
- **ML & Data Framework:** PyTorch, RAPIDS (cuML, cuDF, etc.), scikit-learn
- **Software Framework:** Spring, .NET, Flask
- **Cloud & Deployment:** AWS (EC2, S3, etc), Docker, Terraform, Git
- **Databases:** MySQL, MSSQL, PostgreSQL
- **Operating Systems:** Linux, macOS, Windows

Experience

Graduate Research Assistant

2022 - Present

CompBio-AI Lab, University of North Texas

- Developed advanced AI/ML and deep learning algorithms for genomic data, such as CNNs, graph embeddings, time series, and spatiotemporal models for Hi-C data analysis. Developed scalable data pipelines, algorithmic benchmarking, and deployed web-based tools with Python, Flask, and Docker.

Solutions Developer

2022 – 2022

Otto International

- Developed scalable AWS architecture with Terraform for ESB deployments, optimized Java/PostgreSQL-based services, and established Jenkins-based CI/CD pipelines while working with cross-functional teams to integrate enterprise applications.

Software Engineer

2020 – 2022

Dovetail Technology

- Designed and developed scalable applications using C#, .NET Web API, and MSSQL, deployed using IIS, developed microservices-based Windows services, and worked with other teams to enable the integration of AngularJS and improve workflow.

Software Engineer

2019 – 2020

IdeaScale Bangladesh

- Improvement of backend services with Java, Spring, MySQL, and Docker; backend stability with code refactoring and bug fixes; modernization of legacy SQL with JPA and Querydsl; and working cross-functionally to support the evolution of the scalable system.

Software Engineer

2017 – 2019

Netweaver Software Limited

- Led the development of two e-commerce platforms with Java, Spring Boot, MySQL, SAP UI5, and JavaScript, and a healthcare directory platform with Node.js, Vue.js, and PostgreSQL, focusing on scalable backend development and integration with the frontend.

Awards and Certificates

- MCBIOS 2026 Travel Award
- CENG Anticipated Award, 2025 - University of North Texas
- Graduate School Travel Award, 2024 - University of Colorado Colorado Springs
- Tuition Matching Grant Award, 2022-2024 - University of Colorado Colorado Springs
- Tableau Fundamentals
- Responsible Conduct of Research (RCR-mini)

Leadership and Activities

- Participant, Open House 2025 at UCCS
- President, Bangladeshi Student Association (BSA) 2024-2025 at UCCS
- Participant, Cool Science Festival 2022, 2023, 2024 at UCCS
- Session Chair, NSF REU Summer Research 2023 at UCCS
- Technical Assistant, NSF REU Summer Research 2023 at UCCS
- Conducted a basic Linux and Docker hands-on class in NSF REU Summer Research 2023 at UCCS
- Organized Cumilla Victoria Govt. College Reunion 2016